

Digital Forensics Incident Response Report

Case: "Lockdown" — Web Server Compromise Investigation

Report Type:	Network, Memory & Malware Forensic Analysis
Case Number:	LD-001
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Date of Analysis:	29 August 2026
Client / Scenario:	TechNova Systems (Practice Lab — CyberDefenders)
Evidence Provided:	capture.pcapng (network traffic), memdump.mem (RAM image), updatenow.exe (malware sample)
Tools Used:	Wireshark, Volatility 3, Windows Defender

1. Executive Summary

TechNova Systems' SOC detected suspicious outbound traffic from a public-facing IIS server, suggestive of a web-shell drop and covert command-and-control activity. Forensic analysis of the provided network capture, memory image, and malware sample confirms that the server was compromised via a web-shell (shell.aspx), which was subsequently used to execute a malicious payload identified as Trojan:Win32/AgentTesla.SKAV, a known information-stealing Remote Access Trojan (RAT).

Severity: Critical — confirmed remote code execution and credential-theft malware execution.

2. Attack Timeline

Time (UTC)	Event	Evidence Source
~05:44	Reconnaissance scan from 10.0.2.4 against 10.0.2.15 (Nmap Scripting Engine user-agent detected)	PCAP
~05:44	Failed exploitation attempt via XML-RPC system.listMethods probe (server returned 405)	PCAP
~05:48	Attacker browses /Documents/ directory; only legitimate files present	PCAP
~05:49	shell.aspx (1,015,024 bytes) appears in /Documents/ directory	PCAP
~05:49	Attacker requests /Documents/shell.aspx directly (200 OK)	PCAP
05:44:45	w3wp.exe (IIS worker process) starts, PID 4332	Memory (Volatility)
06:08:23	updatenow.exe (PID 900) spawned as a child of w3wp.exe (PID 4332)	Memory (Volatility)
06:10:48	w3wp.exe process terminates	Memory (Volatility)
N/A	updatenow.exe identified and quarantined as Trojan:Win32/AgentTesla.SKAV	Windows Defender

3. Methodology

- Loaded capture.pcapng in Wireshark and filtered for HTTP traffic to identify the initial reconnaissance and exploitation attempts.
- Identified a User-Agent string matching the Nmap Scripting Engine, confirming automated attacker tooling.

3. Filtered POST requests and reviewed HTTP streams to trace an XML-RPC exploitation attempt that failed (HTTP 405).
4. Reviewed subsequent GET requests to /Documents/ across two timestamps, revealing the appearance of shell.aspx between requests, followed by a direct access to that file.
5. Loaded memdump.mem in Volatility 3, first running windows.info to confirm the memory image's OS version and capture time.
6. Ran windows.pslist to enumerate running processes and their parent-child relationships, identifying updatenow.exe as a child process of the IIS worker process (w3wp.exe).
7. Located the corresponding malware sample and submitted it for local antivirus scanning, which quarantined it and identified the specific malware family.

4. Key Findings

4.1 Initial Access

The attacker (10.0.2.4) used automated tooling (Nmap Scripting Engine) to scan and probe the target IIS server (10.0.2.15). An initial XML-RPC based exploitation attempt was rejected by the server, indicating the attacker likely pivoted to a different method to place a web-shell on the server, as the file appeared shortly afterward without a captured upload mechanism in this traffic sample.

4.2 Web-Shell Deployment and Execution

A file named shell.aspx was placed in the server's /Documents/ directory and subsequently accessed directly by the attacker, resulting in a 200 OK response consistent with successful code execution via the web-shell.

4.3 Malware Execution via Process Injection

Memory analysis confirmed that the IIS worker process (w3wp.exe, PID 4332) spawned a child process, updatenow.exe (PID 900), approximately 24 minutes after the worker process started. This parent-child relationship is strong evidence that the web-shell was used to execute this payload on the compromised server.

4.4 Malware Identification

The executable was identified by endpoint antivirus as Trojan:Win32/AgentTesla.SKAV and automatically quarantined. AgentTesla is a well-documented commodity Remote Access Trojan / keylogger family, known for capturing keystrokes, credentials, and screenshots, and exfiltrating stolen data to an attacker-controlled server.

5. Conclusion

The investigation confirms a full compromise chain: reconnaissance, a web-shell-based initial foothold, and execution of an information-stealing Trojan via the compromised IIS server. The presence of AgentTesla indicates a credential-theft objective, meaning any credentials or sensitive data accessible from this server during the compromise window should be considered at risk.

6. Recommendations

1. Isolate and rebuild the affected server from a known-clean image; do not trust the current OS install.
2. Rotate all credentials that may have been accessible on or from this server (service accounts, admin credentials, stored application secrets).

3. Review IIS logs and file system for the shell.aspx upload mechanism, since the capture in this exercise did not include the initial upload traffic.
4. Deploy application-layer monitoring/WAF rules to detect and block web-shell upload and access patterns.
5. Hunt for other systems that may have communicated with the same attacker infrastructure (10.0.2.4) or exhibit similar w3wp.exe child-process anomalies.

Disclaimer: This report was produced as part of a training exercise using the "Lockdown" practice case provided by CyberDefenders.org. All IP addresses, hostnames, and the scenario described are part of a simulated lab environment and do not represent a real client or organization. This report demonstrates forensic methodology for portfolio purposes.